

SARMELLEK, HungaryWMO: **129220**Lat: **46.683N**Lon: **17.167E**Elev: **410**StdP: **14.48**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	18.2	6.7	7.9	15.9	11.9	10.2	20.9	26.6	34.3	24.2	31.6	7.5	0	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	90.7	71.4	87.8	70.3	84.9	69.1	73.6	86.1	72.0	84.0	70.4	82.1	7.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.6	110.6	80.4	67.9	104.1	78.0	66.3	98.5	76.2	37.4	85.8	35.9	84.0	34.5	82.4	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	20.0	16.8	7.3	95.6	7.1	3.7	2.2	98.2	-1.9	100.3	-5.9	102.4	-11.0	105.1
			6.0	76.7	6.8	1.6	1.1	77.9	-2.9	78.8	-6.7	79.7	-11.7	80.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	32.2	35.7	43.6	53.4	60.3	67.8	70.9	70.5	61.1	52.4	43.2	35.0
	DBStd	15.12	7.92	7.73	8.01	6.38	6.31	6.17	5.59	5.31	5.85	6.88	7.19	7.23
	HDD50	1965	552	400	224	38	4	0	0	0	1	55	222	469
	HDD65	5209	1017	819	664	348	171	42	14	14	145	391	653	931
	CDD50	2793	0	1	24	142	323	534	647	634	334	131	20	3
	CDD65	559	0	0	0	1	24	125	196	183	28	2	0	0
	CDH74	5819	0	0	0	38	312	1288	2019	1836	304	22	0	0
	CDH80	2004	0	0	0	4	53	407	768	704	68	0	0	0
	WSAvg	6.5	6.4	7.3	7.7	7.7	7.2	6.4	6.2	5.6	6.1	5.7	5.9	6.1
	PrecAvg	26.3	1.3	1.3	1.7	2.0	2.3	3.0	3.2	2.8	2.2	1.7	2.8	1.8
(15) Precipitation	PrecMax	40.8	2.5	3.3	4.4	6.2	6.0	5.5	7.5	6.9	4.1	5.6	6.9	3.9
	PrecMin	16.4	0.1	0.3	0.5	0.2	0.1	0.3	0.3	0.5	0.4	0.0	0.8	0.3
	PrecStd	6.2	0.6	0.7	0.9	1.5	1.4	1.4	2.1	1.8	1.1	1.5	1.6	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	60.3	69.8	78.3	84.9	91.7	95.0	95.8	86.5	77.1	66.6	58.7
		MCWB	48.1	49.5	53.9	62.2	65.8	71.4	71.6	70.7	68.5	63.9	57.4	52.3
	2%	DB	50.9	55.1	65.9	74.0	81.0	88.3	90.6	91.0	81.4	72.2	61.1	53.5
		MCWB	45.2	46.7	52.2	58.0	65.4	70.9	71.9	71.0	66.5	62.1	54.7	48.7
	5%	DB	46.8	51.2	61.5	70.8	77.5	85.0	87.6	87.3	77.4	68.6	57.1	48.3
		MCWB	42.6	44.9	50.2	56.6	64.0	69.3	70.4	69.9	64.6	59.9	52.0	44.3
	10%	DB	43.0	46.9	57.1	67.1	74.0	81.7	84.4	83.8	73.4	64.5	53.9	44.5
		MCWB	39.7	42.2	47.8	54.9	61.9	67.6	69.3	68.9	62.7	57.6	50.4	41.4
	0.4%	WB	49.3	50.5	55.1	63.2	69.1	75.0	76.3	75.1	70.0	65.1	58.1	52.7
		MCDB	55.1	59.8	66.4	76.5	80.8	87.3	88.1	87.5	83.1	74.6	65.3	58.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.9	47.5	52.9	59.9	66.9	72.2	73.8	73.3	67.6	62.6	55.2	49.0
		MCDB	50.0	54.1	63.5	71.5	78.2	83.9	86.1	86.3	78.6	71.0	60.2	53.2
	5%	WB	43.0	45.1	50.8	57.5	64.8	70.4	71.9	71.8	65.7	60.5	52.9	44.9
		MCDB	46.2	50.1	60.0	67.7	75.4	82.3	84.2	83.8	75.3	67.4	56.6	47.8
	10%	WB	40.2	42.7	48.7	55.7	62.8	68.8	70.3	70.0	63.6	58.4	50.4	41.8
		MCDB	42.6	46.3	56.2	65.8	72.4	79.9	82.0	81.4	71.4	63.7	53.3	43.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	13.7	18.0	20.5	20.3	20.7	21.3	21.8	19.5	18.4	13.4	11.1
		MCDBR	19.0	21.7	26.2	27.7	26.5	25.7	26.5	28.0	26.2	24.0	19.6	17.2
		MCWBR	13.8	14.3	15.0	14.5	12.9	11.0	10.5	11.1	12.4	14.4	13.4	12.8
		MCDBR	18.2	19.8	24.0	24.6	23.9	23.9	24.3	24.8	23.2	22.9	18.6	16.9
	5% WB	MCWBR	13.6	13.5	14.2	13.2	12.3	10.7	10.8	11.2	11.5	14.1	13.1	12.9
(40) Clear-Sky Solar Irradiance	taub		0.329	0.354	0.389	0.437	0.428	0.431	0.436	0.440	0.418	0.388	0.357	0.329
	taud		2.379	2.324	2.251	2.133	2.212	2.244	2.230	2.230	2.259	2.323	2.362	2.394
	Ebn at Noon		239	255	262	259	266	266	262	256	250	239	224	224
	Edh at Noon		23	30	37	46	44	43	43	41	36	30	23	21
(44) All-Sky Solar Radiation	RadAvg		373	626	1007	1434	1747	1907	1886	1671	1176	773	413	283
	RadStd		47	91	122	141	154	138	133	171	144	83	48	42

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (71 neighbors)		+0.99	N/A	N/A	N/A	+0.86	+0.81	-198	-274	+166	+97